Coding Assignment 3

Due: 20 November, 2026 6pm PT

New homework format

- For this assignment, you will use Google Gemini 4 Hydrogen, available on your UC Davis Google account. Go to `gemini.google.com`, sign in using your UC Davis credentials, and make sure to select the 4 Hydrogen model **without the thinking mode**.
- The submission instructions are completely different from previous coding assignments. Read the entire homework and the instructions carefully.
- **Even if you may have a personal Google account** giving you access to Gemini, you should log into your UC Davis account and use the 4 Hydrogen subscription through the university.
- This coding assignment will also be submitted on Gradescope, but this time as two PDFs (similar to your written homeworks).
- **Sometimes, Gemini crashes.** You can either restart the session, or you can wait for it to run for a while, then close the tab (or Chrome), and reload `gemini.google.com` to find the completed prompt response.
- If Gemini goes "rogue" and produces many pages of slop in a single response, feel free to ignore that session and restart.

Using LLMs Greedily

Task Description

For this assignment, you are asked to interpret and critique the output of Google Gemini 4 Hydrogen (only the base model, **not the thinking model**) on two algorithmic problems. When you graduate and join the workforce, you will need to remain productive, even if you can only use a cheaper Flash-class model, be it due to subscription costs or weekly usage limits.

In the following, you are provided two prompts, containing two algorithmic problems. Copy and paste the prompts exactly to your UC Davis-enabled Gemini 4 **Hydrogen** model. Your task is to interact with the Hydrogen model, and figure out whether Gemini's proposed algorithm is correct or not, and whether the proposed algorithm is in fact efficient enough to run for the desired input size.

In the intended scenario, the **Hydrogen** model will give a correct+efficient algorithm for one of the problems and an incorrect/inefficient one for the other. However, the stochasticity in LLMs means that this might not happen for every student. You should figure out whether your run of the LLM is giving you a correct+efficient algorithm or not, regardless of the "intended" solution. You will be graded on your interaction with the LLM and your final reasoning.

Since this is a new homework format, we will be more generous in grading. The point of this exercise is for you to gain experience critiquing the algorithmic output of LLMs. Note also that such errors can arise even in models more powerful than Flash-class models; however the problems that powerful LLMs get wrong can be a lot more complex, and not suitable for homework use in our course.

Read the submission instruction carefully at the end of this document.

Prompt 1

```
Solve this algorithmic problem exactly. Give the algorithm, a correctness
proof, time complexity, and C++17 code.

There are n files. File i has size a_i. In one operation, you can choose any
two current files of sizes x and y to merge them. The two files are read in
parallel at one unit of data per unit of time; after both reads finish, the
merged file of size x+y is written at the same rate. Thus the single merge
operation takes max(x,y)+x+y time units.

Merge operations can only be performed one at a time, to avoid simultaneous
reads and writes: each merge must finish writing the entire output before
another merge can even start reading files.

Find the minimum total merge time to merge all files into a single file.

Constraints: 1 <= n <= 200000 and 1 <= a_i <= 10^6.
```

Prompt 2

```
Solve this algorithmic problem exactly. Give the algorithm, a correctness
proof, time complexity, and C++17 code.

There are n separate directories. Directory i has nesting depth h_i, defined
as the maximum number of subdirectory levels below it; a directory with no
subdirectories has depth 0.

In one operation, you can choose any two current top-level directories,
create a new directory, and move both chosen directories into it.
Effectively, the two chosen directories are replaced by the new directory,
whose nesting depth is max(x,y)+1 if their depths were x and y.

After exactly n-1 operations, there is one top-level directory left. Find the
minimum possible nesting depth of that final top-level directory.

Constraints: 1 <= n <= 200000 and 0 <= h_i <= 10^9.
```

Submission Instruction

For each problem, select the Google Gemini 4 Hydrogen model (**base model, not the thinking model**) on your UC Davis account, and use the exact copy-and-pasted prompt as the first prompt.

Then, you are free to interact with the model however you want. But, if you blindly write "this is wrong" or something similar, the Hydrogen model seems to second guess itself even if it was correct in the first place. It can even produce a "proof" of NP-hardness after a correct greedy algorithm.

You should end the conversation with one final prompt, of the format:

```
Final answer:

[Some prose to the effect of whether you think the algorithm Gemini gave is
both correct and sufficiently efficient for the stated input size
constraints, and your reasoning]

Do not respond to this prompt.
```

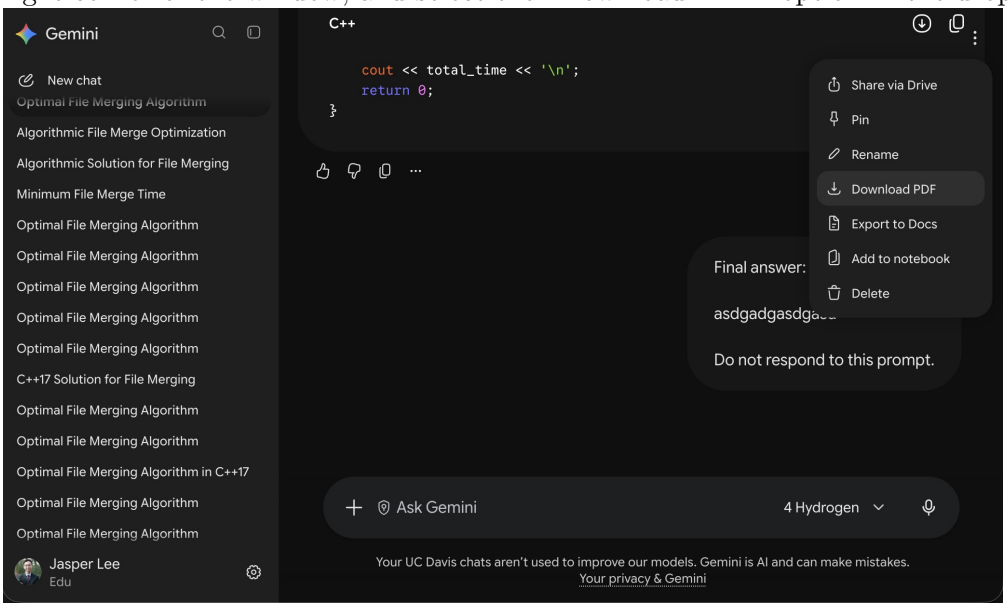
The last line (ideally) prevents Gemini from saying anything further.

We will grade your answer and reasoning against Gemini's last fully presented algorithm (before your "Final answer" message). You may end the chat session whenever you are confident in whether Gemini's algorithm is correct or not, with reasoning to justify that. While you are *allowed* to keep interacting with the model to help you figure out the correctness and efficiency, you do not have to. In fact, you may submit your final answer as the second prompt.

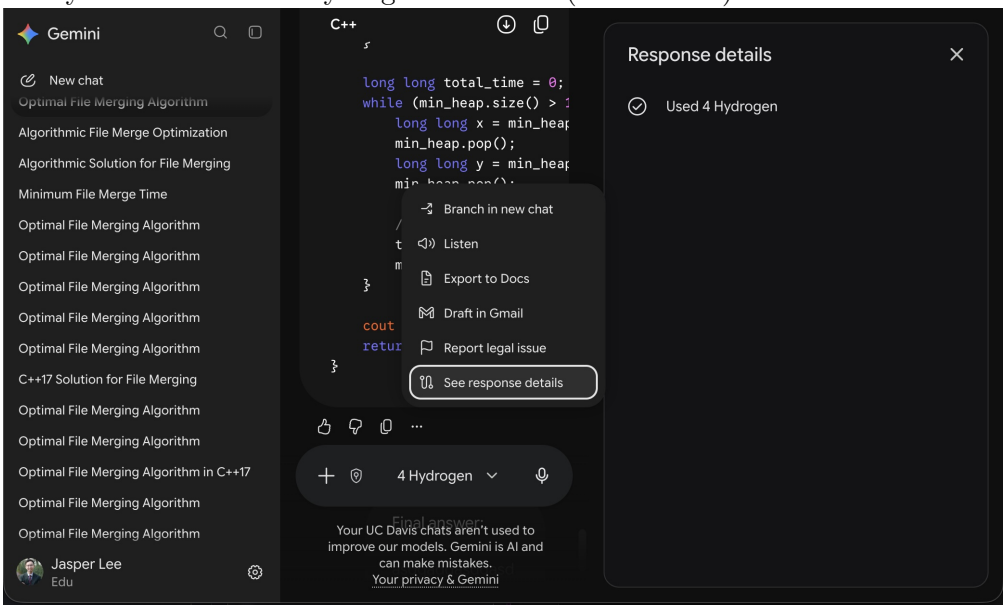
On the other hand, if you choose to further interact with the model, it is ok for you (even encouraged, when done appropriately) to steer Gemini towards a correct algorithm different from its original output. If Gemini mentions something brief/vague about an algorithm you want to be graded against, make sure to prompt it to present the algorithm fully, before you write your final answer.

For each problem, you will start a fresh chat session, and submit the concatenation of two documents.

1. A PDF printout of your entire chat session. You can click the three vertical dots on the top right corner of the window, and select the "Download PDF" option in the dropdown.



2. For the first prompt response (i.e. the first Gemini output answering your copy-and-pasted prompt from this document), click on the horizontal three dots next to the thumbs up/down feedback, and select "See response details" (Gemini sometimes uses a different name for this option). This will show the model used for the first prompt. Take a screenshot, demonstrating that you used the base Hydrogen model for (the start of) this exercise.



Again, use only the base (non-thinking) Gemini 4 Hydrogen model for this assignment. While you could cheat by using a stronger model on the side and we would have no way to tell, we hope that you don't. This new homework is useful for learning only if you engage with the exercise faithfully.